FinanceFirst Bank

Data Center Disaster Recovery Simulation

Course: GRC205 - Incident Management and Business Continuity

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GRC205 Week 21 - Practical Lab Assignment

Data Center Disaster Recovery Simulation

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Executive Summary

FinanceFirst Bank a regional bank with $8.5 billion in assets, serving 250,000 customers across five states (Illinois, Indiana, Wisconsin, Iowa, Michigan), operating in 85 branch locations, employing 1,200 staff, and offering retail banking, commercial lending, wealth management, and mortgage services, recently faced system outage which led to temporary service unavailability across the various channels affecting transaction and 25,000 clients, the outage affected all systems from Core Banking System, Online Banking Portal, Mobile Banking App, ATM Network, Wire Transfer System, Loan Origination System, Email System and Branch Teller system.

The cause of the outage was reported to be a water main break on the street outside the building hosting the primary data center which caused catastrophic flooding at 6:30 am on a Tuesday morning.

Restoration activities were undertaken which included business continuity and disaster recovery team activation, operations from the disaster recovery center and assessment to restore services in the primary data center. While communication was timely and frequently sent, there was panic among clients who overwhelmed the call center and branch management employees, however after media addressed the efforts and focused on restoration activities, trust was restored with noticeable less calls to the bank and more positive reviews.

The bank resumed full operations from the DR center in under 24 hours with all customer-facing systems restored and operational, the CBS fully restored by 5:30 PM same day, wire transfers completed within deadline, ATM network 100% operational at 11:00 PM same day, with minimal data integrity which was resolved with the same night. There were no regulatory violations which was reported, although the system downtime notification was sent out as per requirement, and the media noting positive outcomes. With the operations resumptions, efforts were now redirected towards restoring the primary data center who with the involvement of the insurance team, vendors and internal teams, migration was completed and normalcy resumed in 10 days after initial flooding incident.

The key lessons and takeaway from this incident is how there was disparity between our actual Recovery time objective and estimated and ideal recovery time objective, integrity of the data backup and also the documented recovery and tabletop exercise versus the actual disaster recovery actions and readiness, we also realized importance of planning for eventuality based on lack of availability of a key license, a non-documented technical step and staff fatigue pointing to a gap in skill and training.

Based on the lessons learnt on this incident, our key recommendations include: prioritizing all disaster recovery key licenses to avoid service disruption, always confirm the quality and integrity of our backup data, train staff to avoid gaps in case of fatigue and unavailability, most importantly ensuring our datacenters are flood resilient regardless of location.

Incident Timeline

The below table provides a summary of the timelines of the incident covering initial event through recovery completion:

Action	Key Decision & Actions	System Restoration Milestone	Stakeholder Communication
6:30 AM	Flood at the primary data center	Initial assessment of key asset and extent of damage	Disaster recovery Manager Notification
6:45 AM	Communication to Key stakeholders, activation of the Business continuity and disaster recovery teams	Update on current status, systems affected, decisions on next action steps	Communication to EXCO, Regulators, Branch Managers, IT recovery team,
9:00 AM-12:00 PM	Recovery execution	Disaster recovery plan activated, established and document status of the DR center, system recovery and restoration has begun	Executive Team
12:00-2:00 PM	System assessment and Prioritization, Branch Teller System restored 12:30 Pm, Mobile Banking Restored: 1:30 PM with limited functionality,	Mobile banking operational with limited functionality, Branches now operational, New challenges faced forcing prioritization of the systems and resources, initial primary data center assessment and extent of damage communicated, client and media panic	CEO, EXCO, Media, Regulators, Customer for updates and to reduce panic
2:00 PM- 4:00 PM	Complications and Key Decisions, on staff rotation, CBS operating with missing license, data integrity decision	Recovery in progress with new challenges, Data integrity issues, wire transfer deadline approaching, missing special license key for CBS to run on DR site and staff fatigue	Head of ICT and EXCO to facilitate more resources and escalate CBS Missing key

4:00 PM – 6:00 PM	System Restoration and Confirmation	All customer-facing systems restored, 4:45 PM wire transfers completed beating deadline, 5:30 PM CBS fully operational from the DR.	Communication updates to staff, clients, EXCO, Media and regulators
6:00 PM- 6:00 am	Confirmation of Restoration and next actions	8:00 PM water main repair by city commences 11:00 PM ATM 100% restored Action points for main data center discussed	Insurance, ICT Teams, Business continuity and DR teams
Day 2-10	Primary site recovery	Full systems operations from DR site Efforts and activities to restore primary data and migration of service back to the primary site	EXCO, Staff, Clients, Media, Regulators to align on potential intentional downtime due to migration back to primary site.

Recovery Performance Analysis

In this section we examine and perform root cause analysis for the incident from initial to restoration.

System	Time Restored	Actual RTO	EXPECTED RTO	Difference
Core Banking System	5:30 PM	11 hours	4 hours	7 Hours
Online Banking Portal	1:30 PM	7 hours	2 hours	5 Hours
Mobile Banking App	1:45 PM	7 Hours 15 Minutes	2 hours	5 hours 15 Minutes
ATM Network	11: 00 PM	16 Hours	6 hours	10 Hours
Wire Transfer System	4:15 PM	9 hours 45 Minutes	4 hours	5 Hours 45 Minutes
Loan Origination System	Assumed: 11:PM	16 Hours	24 hours	+ 8 hours
Email System	Assumed: 11:PM	16 Hours	8 hours	8 Hours
Branch Teller System	12:30 PM	6 Hours	4 hours	2 Hours

- **Analysis of what went well**

✓ Team mobilization
✓ site infrastructure is operational
✓ Network connectivity to DR site is stable
✓ Backup data is intact and accessible
✓ Core banking system recovery has begun
✓ Executive team has been briefed and is supportive
✓ DR Center Resilience

- **Analysis of what didn't go well**

❖ Primary data center flooding
❖ Core banking system, and all system recovery is taking longer than expected
❖ Online banking portal requiring manual configuration that wasn't fully documented
❖ 3 key technical staff unavailable
❖ Frequent calls for updates
❖ Customer and media panic
❖ Lack of key license to operate CBS
❖ Data integrity
❖ Staff fatigue

- **Root cause analysis of complications and delays**

Based on the evaluation and analysis of what did not go well, the main root cause of complication and delay is lack of test of the DR center for resilience and actual exercise to validate the documented actions and key steps. Other causes of delay were in resource unavailability including three key staff stuck on traffic, lack of license and missing key steps for some system restorations.

- **Resource utilization and efficiency**

We had sufficient resources however, one key disadvantage was in the skill gap between the technical experts in DR activities and other staff, this caused staff fatigue which called for ad hoc rotations ensuring critical members were divided into groups and assigned as group leads with other team members distributed equally. There was also sufficient coordination among key members with remote assistance facilitated. This was one of the key strong areas that made the disaster recovery efforts smoother.

Decision Analysis

The table below details and reviews major decisions made during the incident and the justification including alternative options and lesson learned.

Key Decision	Justification	Alternative options	Lesson learned
Situation Assessment: decision to address the incident severity as 'Critical'	There was bank-wide system outage affecting all system, the incident affected the primary data center with no certain extent of damage to assets and no clear timeline of recovery, affecting 25, 000 clients and potential loss in finances and reputation	High or Medium, but based on the impact this was a critical incident which needed immediate attention and executive support	How you define an incident affects the response and support granted and also the efforts attributed to resolving the issues, main goal is to understand which incidents are critical and which ones are of less severity.
DR Plan Activation Decision, we chose to activate the DR Plan	All it systems, data and infrastructure was affected with no timelines for restorations calling for DR activation	In this instance there was no alternative options considering the direct implication of not activating the DR plan	Understand when to call for incident response and when to call for full disaster recovery
System Recovery Prioritization, creating an order of restoration	Following customer complaints and pressure from various stakeholders, this decision needed to be made based on the initial RTO as per our initial BIA, change of priorities mid-way will cause confusion. Once the order of priority was created it was easier to even run parallel system restoration which helped in resource allocation and utilization.	Following the pressure and pain points of the shareholders, this would however introduced chaos and divided efficiency, at the same time reduced efforts on main systems	The importance of Business Impact analysis and documentation of RTOs, Importance of following plans as per documentations and not as a reaction to how incidents are perceived.
Resource Allocation: ICT staff rotation	We had 25 staff in total, and the best way was to prioritize for maximum efficiency	Ignore rotation and call for emergency hands-on operations for all ICT staff, however this would have led to fatigue and effort duplication	Importance of having user roles defined and matrix created before incidents. Importance of addressing skill gap by training and drills
Communication	In an incident, one of the key requirements is communication frequently and at correct intervals	Concentrating only on system restoration and focusing on communication after systems are back up,	Decision on who to communicate with and at what interval is very critical.

		however this will only have caused more panic and loss of trust, also executive will not have been available to help if not informed of the situation appropriately	
Data Integrity, choosing which option to go with following reports on corrupt data	We chose option B which meant immediate system restoration using the last latest backup even though it contained 2000 corrupt records which would require to be manually fixed, because it was the most viable option at the time based on the customer and business impact and also on the risk assessment scale,	The alternative to this would either Option A which was taking the last clean back up with 6.5 hours of transaction lost, or delay system restore until data is fixed which was the option which was least viable due to the impact to the customer	The importance of evaluation cost benefit analysis and in our cases number of records affected vs how many clean records we had in hand, also understanding the impact of decisions on the consumer
Wire Transfer Priority	We chose to either attempt parallel recovery of both the CBS and wire transfer system and if it fails seek alternative wire transfer processing method, because this was the option causing the least disruption to other system restoration and to clients	The other alternative options were diverting resources from the core banking system to prioritize wire transfers, which was the least viable option because of the criticality of the CBS, or accept missing wire transfer deadline, this option carried significant potential damages which would have had effects on the bank.	Understand system priorities and options.
Maintain the DR center as the primary data center or return to the primary data center	We chose to start restoration activities and long term plan to return the data center to the primary site due to other dependencies like staff logistics, high operation cost	Maintain the DR center as the main site	Understanding dependencies and effects of decisions on the bigger plan

Stakeholder Management Review

In this section we review how the stakeholder management was coordinated throughout the incident and beyond the incident the measures we take to ensure improved incident response and resilience.

- **Effectiveness of communications**

  From the incident review and analysis, one outstanding element was how effective communication can reduce panic, facilitate stakeholder and executive support and also critical in internal team coordination, from the initial communication to the EXCO and the regulators, media and branch management, the narrative was now controlled by the business continuity team who with continuous communication and frequent update managed the situation.

- **Stakeholder satisfaction assessment**

In a critical incident like this, stakeholder satisfaction assessment must be undertaken in a structure, fast and in relation to our incident. We need to understand the purpose of our assessment ranging from confidence in incident handling procedures, communication feedback, leadership competence and general stakeholder management. We also need to group the stakeholders in various categories so that we have targeted assessment from EXCO to vendor partner and even to the larger staff. We also need an immediate review within 72 hours of the incident and long term review for example 2 or 3 weeks after complete restoration. Once we get the feedback we need to act on the feedback to ensure we perform better in the next incident.

- **Media and public relations outcomes**

In the concluded incident, the media was handled appropriately, with regular notification and update, this also reflected on how restoration was communicated and how the media focused on how quickly we restored the service and did not focus on the outages, and extent of damage.

- **Regulatory compliance and reporting**

The reporting to compliance was also effectively managed with the notification sent to the below regulators. The incident however did not result in any violations.

- Federal Reserve oversight and examination
- FDIC insurance and reporting requirements
- GLBA (Gramm-Leach-Bliley Act) privacy requirements
- Bank Secrecy Act (BSA) and anti-money laundering (AML) compliance
- State banking regulations in five states
- Must notify regulators of significant operational disruptions within 4 hours

Financial Impact Assessment

In this section, we review the financial impact of the incident:

- Direct costs (DR site operation, equipment replacement, overtime)

  There was a high financial impact which was as a result of operating DR site with a cost of $50,000 per day in expenses, additional license key for the CBS, even though equipment replacement was covered by insurance, the labor and resource required to restore the primary site was also cost intensive and including staff overtime for both the system restoration and eventual migration back to the primary site. The financial impact on direct cost was very high.

- Indirect costs (lost revenue, customer attrition, productivity loss)

Post incidence analysis in coordination with finance and recons department have estimated a high financial impact due to the full day system loss, angry clients who switched banks and productivity loss stemming even from indirect costs like ATM incomes.

- Insurance recovery

  The efforts for insurance recovery only covered the damaged equipment and did not cover operational costs.

- Total cost of disaster

  The disaster on initial analysis, based on DR site operations for 10 days, staff overtime, vendor engagement, primary site restoration has been estimated as $ 1, 500, 000.

Lessons Learned

- **What worked well and should be reinforced**

  Based on our analysis, what worked well and should be reinforced and improved: Team mobilization, the coordination and staff rotation, the response to the business continuity activation including call center staff. Site infrastructure is operational, this assisted in quick restoration and the main focus being on data recovery, Network connectivity to DR site is stable, Backup data is intact and accessible with minor corrupt data which was resolved within 24 hours, Executive team that is supportive, DR Center Resilience

- **What didn't work and needs improvement**

  The actual RTO vs expected RTO variance, this can be improved by running more drills, confirming data restoration and DR site operability.

  Review of all systems including the DR site version.

  Documentation confirmation of all critical systems

  Skill gap among staff, affecting resource stability

- **Gaps in DR plan, procedures, or capabilities**

  The DR plan was in place however, more test drill exercise should be done including full operation from the DR site more frequently to avoid last minute critical element loss.

- **Training and preparedness issues identified**

  There were key training gaps identified in the teams, and also lack of proper role and responsibility, it was during the incident response that we were also working on staff rotation, resource mobilization and utilization. There is a need to upskill the key staff members including back up staff to ensure operations are not disrupted due to fatigue or unavailability of key resource.

- **Technology or infrastructure improvements needed**

  One of the main risks in operating from the DR site was lack of key license to operate beyond 24 hours without license, this indicates lack of oversight on backup systems, there is need to have uniform availability of resources between the two sites and priority should be given in addressing uniform and consistent system versions and licenses.

  The other key improvement is on back up procedures to ensure more efficient and resilient data with data integrity and minimal data loss, we need to review the backup procedure and invest in automated backups.

  One other area of improvement is in the data center resilience in both the primary and recovery data centers. This includes more resilience towards natural elements like fire and flood as well as cyber and physical resilience.

Recommendations and Action Plan

Recommendation	Description and Justification	Timelines and milestone	Resource Requirement	Success Metrics
Update disaster recovery plans	Disaster recovery was untested and RTO was not tested for actual expectations and recommendation	Immediate 0- 30 days	Minimal	Disaster recovery plans reviewed and approved
Review of backup process and procedures, regular restoration and test of the backup	One of the key challenges was with data integrity where we had more than 6 hours of data corrupt resulting in manual reconciliation	immediate: review of backup process: 30 days short-term: restoration and test of backup data for correctness to ensure integrity : 90 days long term: Invest in technology that does automatic back up- 6 months	Moderate	Backup data 95% accurate with long term goal of 99.99%
Roles and Responsibilities clarified and clear emergency staff rotation and schedule documented	There was unclear schedules leading to unavailability of key staff as well as fatigue among key members	0-14 days	Minimal	Emergency schedules drafted and acknowledged by all key staff
Incident and disaster recovery communication for call center staff	Call center staff were overwhelmed with calls and require adequate training to ensure seamless response to equiries considering they are critical and customer facing	0-30 days	Minimal	Call center staff improvement levels in customer communication
Critical skill gap training	All critical team members to undergo training in the are of	Long term 6 months	Moderate	Improved incident response

	expertise, there was resource constraint and staff fatigue due to the lack of training and unavailability of critical staff			
Review of critical systems in both the primary and recovery sites	There was a mismatch between licenses in the primary and recovery site, there was also missing documentation on restoration of a key system	Short term for review: 0-30 days Long term for uniformity : 60 days	Moderate	Uniform system features, documented process and procedure
Build a more robust and resilient data center	Due to the gap in resilience review, the primary data center was affected by flooding, to improve resilience the data centers should be confirmed that they are more prepared for any kind of disaster, physical, administrative or technology related	Long term 1 year	High	A more resilient data center with > 90 % score

Appendices

The requirements in this section have been documented in the task response document also included in this assignment submission.